AI PROJECT

Report

Genetic Algorithm Timetable

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Timetable Generation Using Genetic Algorithm

Introduction

The following report discusses the implementation of a genetic algorithm to generate timetables for educational institutions. Timetabling is a complex optimization problem involving the allocation of courses, professors, rooms, and time slots while satisfying various constraints.

Code Overview

The provided code comprises classes and functions for generating timetables using a genetic algorithm. It includes classes such as `TimetableChromosome` for representing timetable solutions and functions for evaluating fitness, selection, crossover, and mutation.

Problem Description

The goal is to create a feasible timetable that allocates courses to sections, professors, rooms, and time slots over a specified number of days. Constraints include avoiding conflicts such as double bookings of rooms or professors, ensuring a balanced workload for professors, and meeting course scheduling requirements.

Genetic Algorithm

The genetic algorithm iteratively evolves a population of timetable solutions to find the optimal timetable. It uses mechanisms inspired by natural selection, including selection, crossover, and mutation, to generate new candidate solutions.

Chromosome Representation

Timetables are represented as chromosomes, with each chromosome encoding the assignment of courses to sections, professors, rooms, and time slots. The chromosome structure is designed to map directly to the timetable variables, facilitating the evaluation and manipulation of solutions.

Fitness Function

The fitness function evaluates the quality of each timetable solution based on predefined constraints. It considers both hard constraints, such as room and professor conflicts, and soft constraints, such as workload balance and course scheduling preferences.

Selection, Crossover, and Mutation

Selection involves choosing parent solutions from the current population based on their fitness scores. Crossover combines genetic material from two parents to produce offspring solutions, while mutation introduces random changes to the offspring's chromosomes to explore new solution space.

Example Usage

An example usage of the code demonstrates how the genetic algorithm generates a feasible timetable solution. The output includes the best fitness score achieved and the corresponding timetable, showing the allocation of courses, sections, professors, rooms, and time slots.

Results and Discussion

The genetic algorithm effectively generates feasible timetables that satisfy the specified constraints. However, the computational complexity of the problem may require further optimization techniques for larger instances. Additionally, fine-tuning the algorithm parameters could improve solution quality and convergence speed.

Conclusion

In conclusion, the implementation of a genetic algorithm for timetable generation demonstrates its effectiveness in solving complex optimization problems. By leveraging evolutionary principles, the algorithm produces high-quality timetables while offering opportunities for further refinement and enhancement.

This report provides insights into the application of genetic algorithms in timetabling problems, highlighting their potential for addressing real-world scheduling challenges in educational institutions.

Output:

```
PS C:\Users\hamza\Desktop\AI Lab> & C:/Users/hamza/AppData/Local/Microsoft/WindowsApps/python3.9.exe "c:/Users/ham
> Generation 0, New Best Fitness: -348
> Generation 0, New Best Fitness: -326
> Generation 0, New Best Fitness: -310
> Generation 0, New Best Fitness: -302
Early stopping at generation 0
Best Fitness: -302
Best Timetable:
Day Monday, Time 8:30 AM - 9:50 AM, Room 0, Course Accounting, Section Section 3, Professor Professor D
Day Monday, Time 8:30 AM - 9:50 AM, Room 4, Course Physics, Section Section 3, Professor Professor C
Day Monday, Time 8:30 AM - 9:50 AM, Room 5, Course Physics, Section Section 1, Professor Professor C
Day Monday, Time 10:00 AM - 11:20 AM, Room 6, Course Accounting, Section Section 5, Professor Professor D
Day Monday, Time 11:30 AM - 12:50 PM, Room 3, Course Accounting, Section Section 1, Professor Professor D
Day Monday, Time 11:30 AM - 12:50 PM, Room 5, Course Physics, Section Section 4, Professor Professor C
Day Monday, Time 1:00 PM - 2:20 PM, Room 2, Course Accounting, Section Section 4, Professor Professor D
Day Monday, Time 2:30 PM - 3:50 PM, Room 4, Course English, Section Section 2, Professor Professor E
Day Monday, Time 2:30 PM - 3:50 PM, Room 9, Course Maths, Section Section 3, Professor Professor B
Day Monday, Time 4:00 PM - 5:20 PM, Room 3, Course Physics, Section Section 5, Professor Professor C
Day Monday, Time 4:00 PM - 5:20 PM, Room 5, Course Physics, Section Section 2, Professor Professor C
Day Monday, Time 4:00 PM - 5:20 PM, Room 9, Course Programming, Section Section 2, Professor Professor A
Day Tuesday, Time 8:30 AM - 9:50 AM, Room 2, Course Programming, Section Section 4, Professor Professor A
Day Tuesday, Time 8:30 AM - 9:50 AM, Room 9, Course English, Section Section 1, Professor Professor E
Day Tuesday, Time 10:00 AM - 11:20 AM, Room 1, Course Programming, Section Section 3, Professor Professor A
Day Tuesday, Time 11:30 AM - 12:50 PM, Room 4, Course Programming, Section Section 1, Professor Professor A
Day Tuesday, Time 11:30 AM - 12:50 PM, Room 6, Course English, Section Section 2, Professor Professor E
Day Tuesday, Time 1:00 PM - 2:20 PM, Room 6, Course Accounting, Section Section 1, Professor Professor D
Day Tuesday, Time 2:30 PM - 3:50 PM, Room 0, Course Programming, Section Section 1, Professor Professor A
Day Tuesday, Time 4:00 PM - 5:20 PM, Room 3, Course Physics, Section Section 2, Professor Professor C
Day Tuesday, Time 4:00 PM - 5:20 PM, Room 7, Course Physics, Section Section 5, Professor Professor C
Day Wednesday, Time 8:30 AM - 9:50 AM, Room 8, Course Maths, Section Section 2, Professor Professor B
Day Wednesday, Time 10:00 AM - 11:20 AM, Room 8, Course Accounting, Section Section 3, Professor Professor D
Day Wednesday, Time 11:30 AM - 12:50 PM, Room 0, Course English, Section Section 3, Professor Professor E
Day Wednesday, Time 1:00 PM - 2:20 PM, Room 2, Course English, Section Section 5, Professor Professor E
Day Wednesday, Time 1:00 PM - 2:20 PM, Room 6, Course Programming, Section Section 5, Professor Professor A
Day Wednesday, Time 2:30 PM - 3:50 PM, Room 5, Course Physics, Section Section 1, Professor Professor C
Day Wednesday, Time 2:30 PM - 3:50 PM, Room 6, Course Accounting, Section Section 2, Professor Professor D
Day Wednesday, Time 2:30 PM - 3:50 PM, Room 8, Course Accounting, Section Section 5, Professor Professor D
Day Wednesday, Time 4:00 PM - 5:20 PM, Room 2, Course Maths, Section Section 2, Professor Professor B
Day Thursday, Time 10:00 AM - 11:20 AM, Room 7, Course Physics, Section Section 3, Professor Professor C
Day Thursday, Time 10:00 AM - 11:20 AM, Room 8, Course Programming, Section Section 2, Professor Professor A
Day Thursday, Time 11:30 AM - 12:50 PM, Room 1, Course Programming, Section Section 4, Professor Professor A
Day Thursday, Time 11:30 AM - 12:50 PM, Room 2, Course Maths, Section Section 1, Professor Professor B
Day Thursday, Time 1:00 PM - 2:20 PM, Room 0, Course English, Section Section 4, Professor Professor E
Day Thursday, Time 1:00 PM - 2:20 PM, Room 5, Course Programming, Section Section 5, Professor Professor A
Day Thursday, Time 2:30 PM - 3:50 PM, Room 5, Course Maths, Section Section 5, Professor Professor B
Day Thursday, Time 2:30 PM - 3:50 PM, Room 9, Course English, Section Section 1, Professor Professor E
Day Thursday, Time 4:00 PM - 5:20 PM, Room 1, Course Maths, Section Section 1, Professor Professor B
Day Friday, Time 8:30 AM - 9:50 AM, Room 2, Course Maths, Section Section 3, Professor Professor B
Day Friday, Time 8:30 AM - 9:50 AM, Room 3, Course Maths, Section Section 5, Professor Professor B
Day Friday, Time 10:00 AM - 11:20 AM, Room 5, Course Programming, Section Section 3, Professor Professor A
Day Friday, Time 11:30 AM - 12:50 PM, Room 0, Course English, Section Section 4, Professor Professor E
Day Friday, Time 11:30 AM - 12:50 PM, Room 5, Course Maths, Section Section 4, Professor Professor B
Day Friday, Time 11:30 AM - 12:50 PM, Room 9, Course Accounting, Section Section 2, Professor Professor D
Day Friday, Time 1:00 PM - 2:20 PM, Room 6, Course English, Section Section 3, Professor Professor E
Day Friday, Time 2:30 PM - 3:50 PM, Room 0, Course Accounting, Section Section 4, Professor Professor D
Day Friday, Time 2:30 PM - 3:50 PM, Room 4, Course English, Section Section 5, Professor Professor E
Day Friday, Time 2:30 PM - 3:50 PM, Room 9, Course Maths, Section Section 4, Professor Professor B
Day Friday, Time 4:00 PM - 5:20 PM, Room 0, Course Physics, Section Section 4, Professor Professor C
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